WORKED SOLUTION

STEP II 1990, Question 12

Mechanics • Collision

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Read the question before you go on. It is in the STEP Support Programme's question database:

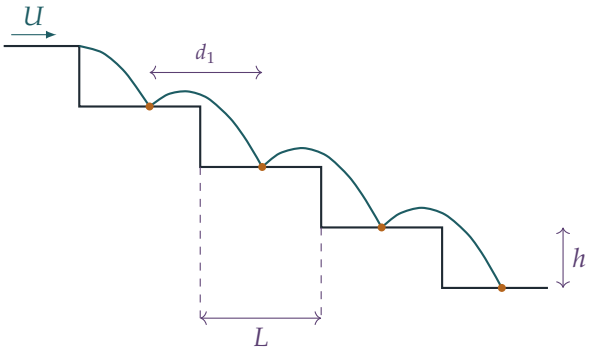
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The key idea

The stairs are smooth, so the horizontal velocity never changes: it stays U throughout, and every horizontal distance is just U times a flight time. All the work is in the vertical motion, where each bounce scales the vertical speed by e and each drop of h adds $2gh$ to its square. That gives a simple recurrence for the *square* of the vertical speed at each bounce. For the last part, think about where the bounces must land: one per stair, on a staircase that goes on for ever.

Solution

We model the particle as a point moving under gravity alone between bounces, and the stairs as smooth horizontal surfaces, each a height h below the last, with edges a horizontal distance L apart.



The distance between bounces

Horizontal motion. The stairs are smooth, so each impulse from a stair is purely vertical, and between bounces the only force is gravity. So the horizontal velocity is U

at all times, and a flight lasting time T covers horizontal distance UT.

Vertical motion. Let w_n be the speed (downwards) with which the particle hits the stair at its nth bounce. It leaves the top stair with no vertical velocity and falls h, so

$$w_1^2 = 2gh.$$

At the nth bounce, Newton's law of restitution with a fixed stair sends it upwards at speed ew_n. It then has to end up h lower, on the next stair, so by $v^2 = u^2 + 2as$ (taking downwards as positive, the displacement is $+h$):

$$w_{n+1}^2 = e^2 w_n^2 + 2gh.$$

Starting from $w_1^2 = 2gh$ and applying this repeatedly,

$$\begin{aligned} w_n^2 &= 2gh\left(1 + e^2 + e^4 + \cdots + e^{2(n-1)}\right) \\ &= 2gh\,\frac{1 - e^{2n}}{1 - e^2} = 2gh\,\alpha_n^2, \end{aligned}$$

summing the geometric series (valid as $e^2 \neq 1$). So $w_n = \sqrt{2gh}\,\alpha_n$ for every $n \geq 1$; a quick induction makes this formal, since $e^2 \cdot 2gh\,\alpha_n^2 + 2gh = 2gh\left(e^2\alpha_n^2 + 1\right)$ and $e^2\alpha_n^2 + 1 = \dfrac{e^2 - e^{2n+2} + 1 - e^2}{1 - e^2} = \alpha_{n+1}^2$.

Ask yourself How long is the flight from the nth bounce to the $(n+1)$th?

The vertical velocity changes at the constant rate g (downwards). It starts at ew_n upwards and finishes at w_{n+1} downwards, so it changes by $ew_n + w_{n+1}$ in total, and the flight time is

$$T_n = \frac{ew_n + w_{n+1}}{g} = \frac{\sqrt{2gh}}{g}\left(e\alpha_n + \alpha_{n+1}\right) = \left(\frac{2h}{g}\right)^{\frac{1}{2}}\left(e\alpha_n + \alpha_{n+1}\right).$$

(This avoids solving a quadratic in T_n: we already know the final speed.) Multiplying by the horizontal speed U,

$$d_n = U\left(\frac{2h}{g}\right)^{\frac{1}{2}}\left(e\alpha_n + \alpha_{n+1}\right),$$

as required.

A very long staircase

Since $0 \leq e < 1$, we have $e^{2n} \to 0$, so $\alpha_n \to \dfrac{1}{\sqrt{1 - e^2}}$ as $n \to \infty$, and therefore

$$d_n \to D = U\left(\frac{2h}{g}\right)^{\frac{1}{2}} \frac{1 + e}{\sqrt{1 - e^2}} = U\left(\frac{2h}{g}\right)^{\frac{1}{2}}\left(\frac{1 + e}{1 - e}\right)^{\frac{1}{2}},$$

using $1 - e^2 = (1 - e)(1 + e)$.

Ask yourself Why must this limiting distance be exactly L?

Measure x horizontally from the edge of the top stair. The first stair below occupies $0 \le x \le L$, and in general the nth stair below the top occupies $(n-1)L \le x \le nL$. The particle bounces once on every stair, so its nth bounce, at x_n say, satisfies

$$(n-1)L \le x_n \le nL, \qquad \text{so} \qquad \frac{x_n}{n} \to L \text{ as } n \to \infty.$$

On the other hand, $x_n = x_1 + d_1 + d_2 + \cdots + d_{n-1}$, and $d_k \to D$. If $D > L$, then from some point on every d_k exceeds L by at least a fixed amount, so the particle gains a fixed distance on the stair edges with every bounce and eventually overshoots a stair; if $D < L$ it falls steadily behind and eventually lands twice on the same stair. (Formally, the average x_n/n of terms tending to D tends to D.) Either way, a long enough staircase is not bounced on once per stair. So for very large N we need $D = L$:

$$U\left(\frac{2h}{g}\right)^{\frac{1}{2}}\left(\frac{1+e}{1-e}\right)^{\frac{1}{2}} = L \quad \Longrightarrow \quad U = \left(\frac{L^2 g}{2h}\right)^{\frac{1}{2}}\left(\frac{1-e}{1+e}\right)^{\frac{1}{2}},$$

as required.

Checks

- *Special values.* $\alpha_1 = 1$, so $d_1 = U\sqrt{2h/g}\,(e + \sqrt{1+e^2})$. With $e = 0$ the particle stops dead vertically at each bounce, so each flight is a plain drop of h taking $\sqrt{2h/g}$: indeed $d_n = U\sqrt{2h/g}$, and the final condition becomes $U\sqrt{2h/g} = L$, exactly as it should.
- *Dimensions.* $\sqrt{2h/g}$ is a time, so d_n is a length; $\sqrt{L^2 g/h}$ is a speed.
- *Numerically.* Stepping the motion numerically for random values of U, h and e (with $g = 9.8$) reproduced each d_n to within 10^{-6}. With $e = \frac{1}{2}$, $L = 0.3$ and $h = 0.2$, the formula gives $U \approx 0.857$, and the bounce positions then stayed on successive stairs for 2000 stairs.

Common slips

- Taking every bounce to be a drop of h from rest, or from the height of the previous bounce: the particle rises after each bounce, so its vertical speed builds up from stair to stair.
- Writing $w_{n+1} = ew_n + \sqrt{2gh}$: it is the *squares* of the speeds that add, not the speeds themselves.

Beyond STEP

The last part is a statement about averages: if $d_n \to D$ then the running mean $\frac{1}{n}\sum d_k \to D$ too (Cesàro's theorem), which is a standard early exercise in real analysis. Notice too that the condition on U is *necessary*, not sufficient: the early bounces are shorter than L, and if e is close to 1 the accumulated shortfall can exceed the head start x_1, so no launch speed works at all. For limits of sequences done carefully, see Abbott, *Understanding Analysis*; for the mechanics of impacts, Kleppner and Kolenkow, *An Introduction to Mechanics*.

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